DUAL NATURE OF MATTER

Dual Nature

Definition:

Radiation and matter show both wave nature and particle nature. This is called dual nature.

Photoelectric Effect

Definition:

The phenomenon of emission of electrons from metal surface when light falls on it is called photoelectric effect.

Photoelectrons:

The emitted electrons are called photoelectrons.

Einstein's Photoelectric Equation

$h\nu = \phi + KE$

Where:

- h = Planck's constant
- ν = Frequency of light
- ϕ = Work function

- KE = Kinetic energy

Work Function

Minimum energy required to remove an electron from a metal surface is called work function.

Threshold Frequency

Definition:

The minimum frequency of incident radiation required to produce photoelectric effect is called threshold frequency.

Formula:

$\phi = h\nu_0$

Where:

- ϕ = Work function
- h = Planck's constant
- ν_0 = Threshold frequency

de Broglie Hypothesis

Statement:

According to de Broglie, moving particles are associated with waves.

de Broglie Wavelength:

$\lambda = h / p$

Where:

- λ = Wavelength
- h = Planck's constant
- p = Momentum

Davisson-Germer Experiment

This experiment proved the wave nature of electrons.

Matter Waves

Definition:

The waves associated with moving particles are called matter waves.

Properties:

- Matter waves are not electromagnetic waves.
- Matter waves are associated with moving particles.
- Wavelength depends on momentum.
- Smaller the momentum, larger the wavelength.

Heisenberg’s Uncertainty Principle

Statement:

It is impossible to determine simultaneously the exact position and momentum of a particle.

Formula:

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

Where:

- Δx = Uncertainty in position
- Δp = Uncertainty in momentum
- h = Planck’s constant

Applications

- Electron microscope

● Study of atomic particles

● Quantum mechanics

Quick Revision Points

● Radiation and matter show dual nature.

● The photoelectric effect proves the particle nature of light.

● Einstein explained the photoelectric effect.

● de Broglie proposed the wave nature of particles.

● Matter waves are associated with moving particles.

● The Davisson-Germer experiment proved the wave nature of electrons.

● Exact position and momentum cannot be measured simultaneously.

Important Formulae

$$h\nu = \phi + KE$$

$$\phi = h\nu_0$$

$$\lambda = h / p$$

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

Conclusion

The dual nature of matter and radiation forms the foundation of modern quantum physics.